

Powers with a natural exponent

Repeated multiplication, written short — and the vocabulary that goes with it.

LESSON 61–62

2 × 40 MIN

MATEMATIKA 7, §27

S8 6.1

LESSON CLOCK

15'

25'

25'

20'

5'

The definition	15 min
Signs and fractions	25 min
Order of operations	25 min
Squares and cubes	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define a^n for a natural exponent and name the base and the exponent.
- Compute powers of positive, negative and fractional bases.
- Use the correct order of operations with powers.
- Recognise squares and cubes of the small numbers.

TEXTBOOK

National	pp. 164–170
Cambridge	Stage 8, pp. 58–62
Workbook	Exercise 6.1

01

Explanation

The definition

$$a^n = a \times a \times \dots \times a \text{ (} n \text{ factors), } n \in \mathbb{N}$$

a is the **base**, n the **exponent**. $a^1 = a$, and a^2 is read “ a squared”, a^3 “ a cubed”.

POWER	MEANS	VALUE
2^5	$2 \times 2 \times 2 \times 2 \times 2$	32
3^4	$3 \times 3 \times 3 \times 3$	81
10^3	$10 \times 10 \times 10$	1000
7^1	7	7

2⁵ IS NOT 2 × 5

It is 32, not 10. The exponent counts the factors; it is not a multiplier. Reading the power aloud as “two multiplied by itself five times over” prevents the confusion.

Signs and fractions

POWER	VALUE	WHY
$(-2)^3$	-8	three negative factors
$(-2)^4$	16	four negative factors
-2^4	-16	only the 2 is raised
$(\frac{2}{3})^3$	$\frac{8}{27}$	numerator and denominator both
$(0.1)^2$	0.01	0.1×0.1

$$(-a)^n = a^n \text{ for even } n, (-a)^n = -a^n \text{ for odd } n$$

THE BRACKETS DECIDE WHAT IS RAISED

$(-2)^4 = 16$ but $-2^4 = -16$. This distinction is tested every year from now to Grade 11.

Order of operations

Powers come after brackets and before multiplication.

EXPRESSION	WORKING	VALUE
3×2^3	3×8	24
$(3 \times 2)^3$	6^3	216
$2 + 3^2$	$2 + 9$	11
$(2 + 3)^2$	5^2	25
$5^2 - 3 \times 2^2$	$25 - 12$	13

A BRACKET IS THE ONLY WAY TO RAISE A SUM

$(2 + 3)^2$ is 25; $2 + 3^2$ is 11; and $2^2 + 3^2$ is 13. Three different expressions, and the brackets are what distinguish them.

Squares and cubes

N	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Worth adding: $2^5 = 32$, $2^6 = 64$, $2^7 = 128$, $2^8 = 256$, $2^{10} = 1024$, and $3^4 = 81$, $5^4 = 625$.

THESE TWO ROWS SAVE MORE TIME THAN ANY RULE

They are used in factorising, in simplifying surds, in Pythagoras and in every prime factorisation. Learning them once repays the effort for five years.

02

Worked examples

EXAMPLE 1 Compute $(-2)^4$, -2^4 and $(-2)^5$.

1

$(-2)^4$: four negative factors — positive.

2

$= 16$

3

-2^4 means $-(2^4)$ — only the 2 is raised.

4

$= -16$; and $(-2)^5 = -32$.
Five factors — odd.

ANSWER

16, -16, -32

EXAMPLE 2 Compute $5^2 - 3 \times 2^2$.

- ① Powers first: 25 and 4.
- ② Then multiplication: $3 \times 4 = 12$.
- ③ Then subtraction.
- ④ $25 - 12 = 13$

ANSWER

13

EXAMPLE 3 Compute $(\frac{2}{3})^3$ and $(0.1)^2$.

- ① Raise the numerator and the denominator.
- ② $\frac{2^3}{3^3} = \frac{8}{27}$
- ③ 0.1×0.1
- ④ $= 0.01$

ANSWER

 $\frac{8}{27}$ and 0.01**03 Interactive model**

Fold a sheet of paper in half repeatedly and count the layers; the class reaches 2^7 before the paper stops folding, and the growth is felt rather than stated.

Powers of a number

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Power	Daraja	Степень
Base	Asos	Основание
Exponent	Ko'rsatkich	Показатель
Square	Kvadrat	Квадрат
Cube	Kub	Куб
To raise to a power	Darajaga ko'tarish	Возвести в степень
Repeated multiplication	Takroriy ko'paytirish	Повторное умножение
Order of operations	Amallar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** 2^5 equals:In a^n , n is the: $(-2)^4$ equals:

-2^4 equals:

16

-16

8

-8

$(2 + 3)^2$ equals:

11

13

25

36

$(\frac{2}{3})^3$ equals:

$\frac{2}{27}$

$\frac{8}{27}$

$\frac{6}{9}$

$\frac{8}{3}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 2^5

ANSWER 32

2. 3^4

ANSWER 81

3. 10^3

ANSWER 1000

4. 7^2

ANSWER 49

5. 5^3

ANSWER 125

6. $(-2)^3$

ANSWER -8

7. $(-2)^4$

ANSWER 16

Medium · the standard question

7 PROBLEMS

1. -2^4

ANSWER -16

2. $(\frac{2}{3})^3$

ANSWER $\frac{8}{27}$

3. $(0.1)^2$

ANSWER 0.01

4. 3×2^3

ANSWER 24

5. $(3 \times 2)^3$

ANSWER 216

6. $2 + 3^2$

ANSWER 11

7. $5^2 - 3 \times 2^2$

ANSWER 13 **Hard · stretch and reason**

7 PROBLEMS

1. $(2 + 3)^2 - 2^2 - 3^2$

ANSWER 12

2. $(-3)^2 + (-3)^3$

ANSWER -18

3. $2^8 \div 2^5$

ANSWER 8

4. $(0.2)^3$

ANSWER 0.008

5. $(-\frac{1}{2})^4$

ANSWER $\frac{1}{16}$

6. Which is larger, 2^{10} or 10^3 ?

ANSWER $2^{10} = 1024$

7. $3^2 \times 3^3$

ANSWER 243

07 Homework**Homework — 5 tasks**

Write out the factors for any power you are unsure of.

1. Compute 2^6 , 3^5 and 10^4 .

2. Compute $(-3)^3$, $(-3)^4$ and -3^4 .

3. Compute $(\frac{3}{4})^2$ and $(0.3)^2$.

4. Compute $4^2 + 2 \times 3^2$.

5. Learn the squares to 12 and the cubes to 6, and write them out from memory.